

Technical Decision Records (TDR / ADR)

Project: LearnLens AI

ADR-001: Universal Browser `getDisplayMedia` vs. Platform-Specific Scrapers

- **Status:** Accepted
 - **Context:** The product must observe courses across Coursera, YouTube, Fortinet, Forage, Udemy, local video files, and private university portals.
 - **Decision:** Utilize the browser's standard `navigator.mediaDevices.getDisplayMedia` capture API rather than writing individual web scrapers or browser bots for each platform.
 - **Consequences:**
 - **Pros:** 100% platform-independent; zero vulnerability to website HTML/DOM layout changes; zero violation of platform terms of service; works on authenticated enterprise intranets.
 - **Cons:** Requires explicit user consent via the browser picker (which reinforces user privacy).
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ADR-002: Perceptual Difference Hashing (dHash) for Frame Deduplication

- **Status:** Accepted
 - **Context:** Educational video lectures display static slides for 30–120 seconds. Taking screenshots blindly every 2 seconds creates thousands of redundant images, exhausting disk space and CPU.
 - **Decision:** Implement a 64-bit difference hash (dHash) using OpenCV and calculate Hamming distance against prior frames. Only frames with Hamming distance ≥ 8 are processed.
 - **Consequences:** Reduces downstream OCR, VLM, and disk operations by over 85%, allowing smooth operation on ordinary student laptops.
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ADR-003: Session-Scoped Knowledge Namespaces vs. Monolithic Global RAG

- **Status:** Accepted
 - **Context:** A student studying Network Firewalls on Fortinet today should not receive answers contaminated by an Art History course they studied yesterday.
 - **Decision:** Partition vector indices and SQLite records strictly per `session\_id`. Cross-session search is only performed when the user explicitly requests "Search all my learning".
 - **Consequences:** Completely eliminates inter-domain hallucination and keeps vector lookups under 5 milliseconds.
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ADR-004: Ethical Assessment Boundary vs. Automated Quiz Botting

- **Status:** Accepted
 - **Context:** Some users might request automated quiz submission to pass certifications.
 - **Decision:** Implement formative assessment features (importing quiz PDFs, diagnosing weak concepts, explaining why an answer is correct, creating mock quizzes) while strictly prohibiting automated exam submission or credential harvesting.
 - **Consequences:** Maintains ethical academic integrity while maximizing human learning outcomes.
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ADR-005: Local Python Presentation & PDF Generators (`python-pptx`, `reportlab`)

- **Status:** Accepted
- **Context:** Users require downloadable PowerPoint presentations and study guide PDFs.
- **Decision:** Use `python-pptx` to construct native widescreen presentation decks embedding real captured diagrams, and `reportlab` to compile structured study reports with tables and callout boxes.
- **Consequences:** Zero dependency on paid cloud document APIs; instant local downloads with consistent typography.